How processing energy is calculated

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What the model calculates

Processing energy per kg of food = the energy of every process on the way from raw commodity to food, each scaled to how much raw material that process handles for 1 kg of food. Two files hold the inputs:

- **Recipe file** (SAFAD IP Recipes.csv): what each food is made of (share per kg), the process on each ingredient row (facet), the **reverse yield** (kg raw material per kg product) and the **allocation factor** (share of the raw material's burden given to this product). It also holds one **preparation process** per food (for example baking for bread) with its own preparation reverse yield.
- **Energy file** (SAFAD IP Energy Proc.csv): one value per process (facet), in **MJ final energy per kg raw material entering the process**, split into energy carriers (electricity, natural gas and so on).

A food's processing energy is the energy of all its ingredient rows (each ingredient's own recipe included, level by level) plus its own preparation step, counted once. For food f with ingredient rows i:

$$E_{\mathrm{f}} = \Sigma_{\mathrm{i}}\, s_{\mathrm{i}} \cdot p_{\mathrm{i}} \cdot RY_{\mathrm{i}} \cdot A_{\mathrm{i}} \cdot \left(\Sigma_{\mathrm{k}}\, C_{\mathrm{k,i}} + E^{*}_{\mathrm{i}} \right) \; + \; C_{\mathrm{prep,f}} \cdot M_{\mathrm{f}}$$

$$E^{*}_{\mathrm{i}} = \begin{cases} 0 & \text{raw commodity (no recipe)} \\ E_{\mathrm{i}} & \text{if } N_{\mathrm{i}} = 1 \\ E_{\mathrm{i}} - C_{\mathrm{prep,i}} \cdot M_{\mathrm{i}} & \text{if } N_{\mathrm{i}} = 0 \end{cases}$$

s = share (Percentage / 100), p = probability, RY = reverse yield, A = allocation factor, C_k = energy file value of each process k on the row (facet), E_i = the ingredient's own processing energy from the same equation (one level down), C_prep × M = the food's preparation step (columns K–M), N = column "Include ingredient's preparation (1/0)".

Example 1: Wheat bread, white

1 kg of white wheat bread (A.01.04.001.001) needs **3.62 MJ** of processing energy, 90 % of it for baking. The recipe describes the finished bread, so the shares add up to 1 kg of bread (the water that evaporates in the oven is not in the

recipe).

Ingredient	kg per kg bread	Process on the row	Where its energy comes from
Wheat flour, white	0.7429	None (its own recipe)	flour recipe: wheat grain, Grain milling
Tap water	0.2000	Unspecified	none
Sunflower oil	0.0286	None (its own recipe)	oil recipe: sunflower seed, Oil production
Salt	0.0171	Unspecified	none
Yeast	0.0114	Unspecified	none
Bread itself	**1**	**Preparation: Roasting (baking), preparation reverse yield 1.12**	**baking the dough**

How to read the recipe extracts: columns A–N as in tab Recipes of the recipe file. Grey = same value as the row above (the food and its preparation repeat on every row). **Yellow** = reverse yield or allocation other than 1. **Orange 0** in column N = the ingredient's own preparation is left out.

The same recipe as it stands in the recipe file (tab Recipes), all columns. The preparation columns repeat on every row, but the model counts the preparation once per food.

FOOD		INGREDIENT (COMPONENT)		PROCESS ON THE ROW		AMOUNT AND FACTORS				FOOD'S PREPARATION (SAME ON EVERY ROW, COUNTED ONCE)			
A **Food code**	B **Food name**	C **Component code**	D **Component name**	E **Facet(s) code**	F **Facet(s) Description**	G **Percentage**	H **Probability**	I **Reverse Yield Factor**	J **Allocation Factor**	K **Preparation process code**	L **Preparation process**	M **Preparation reverse yield**	N **Include ingredient's preparation (1/0)**
A.01.04.001.001	Wheat bread, white	A.01.03.001.003	Wheat flour, white	To be further disaggregated (d)		74.29	1	1	1	F28.A07GY	Roasting	1.12	1
A.01.04.001.001	Wheat bread, white	A.11.04.017	Sunflower oil	To be further disaggregated (d)		2.86	1	1	1	F28.A07GY	Roasting	1.12	1
A.01.04.001.001	Wheat bread, white	A.15.01	Tap water	F28.A07XD	PROCESS=Unspecified	20	1	1	1	F28.A07GY	Roasting	1.12	1
A.01.04.001.001	Wheat bread, white	A.16.04.001	Salt	F28.A07XD	PROCESS=Unspecified	1.71	1	1	1	F28.A07GY	Roasting	1.12	1
A.01.04.001.001	Wheat bread, white	A.16.10.001	Yeast	F28.A07XD	PROCESS=Unspecified	1.14	1	1	1	F28.A07GY	Roasting	1.12	1

Step 1: energy for processing the ingredients

The ingredients add **0.343 MJ** per kg bread: 0.284 MJ from milling the flour and 0.059 MJ from producing the oil. Each ingredient is first calculated per kg of that ingredient from its own recipe: its processing energy in MJ per kg flour or per kg oil. That value is then multiplied by the ingredient's share in the bread (kg per kg bread).

Flour. The flour recipe is wheat grain with the process Grain milling. The preparation columns are **empty**: flour has no preparation process since it is an ingredient, so its only energy is the grain milling on this row.

FOOD		INGREDIENT (COMPONENT)		PROCESS ON THE ROW		AMOUNT AND FACTORS				FOOD'S PREPARATION (SAME ON EVERY ROW, COUNTED ONCE)			
A **Food code**	B **Food name**	C **Component code**	D **Component name**	E **Facet(s) code**	F **Facet(s) Description**	G **Percentage**	H **Probability**	I **Reverse Yield Factor**	J **Allocation Factor**	K **Preparation process code**	L **Preparation process**	M **Preparation reverse yield**	N **Include ingredient's preparation (1/0)**
A.01.03.001.003	Wheat flour, white	A.01.02.001	Wheat grain	F28.A0C03$ F02.A067Z$ F10.A07XK	PROCESS=Grain milling, PART-NATURE=Flour/meal or finely ground powder (as part-nature), QUALITATIVE-INFO=White/refined	100	1	**1.43**	**0.86**				1

1. Energy file: Grain milling = **0.311 MJ per kg wheat grain**.
2. Reverse yield **1.43**: 1 kg white flour is made of 1.43 kg grain.
3. Allocation **0.86**: the flour carries 86 % of the grain's burden; bran and germ and feed carry the rest (economic allocation). Feed = middlings and feed flour, about 0.10 kg per kg grain, sold as animal feed; it carries about 4 % (bran 7 %, germ 2.5 %).
4. Per kg flour: 0.311 × 1.43 × 0.86 = **0.383 MJ**.
5. In the bread: 0.7429 kg flour × 0.383 = **0.284 MJ**.

Sunflower oil. The oil recipe is sunflower seed with the process Oil production.

The oil recipe as it stands in the recipe file (one row). The preparation columns are empty: the oil's only energy is the oil production on this row.

FOOD		INGREDIENT (COMPONENT)		PROCESS ON THE ROW		AMOUNT AND FACTORS				FOOD'S PREPARATION (SAME ON EVERY ROW, COUNTED ONCE)			
A **Food code**	B **Food name**	C **Component code**	D **Component name**	E **Facet(s) code**	F **Facet(s) Description**	G **Percentage**	H **Probability**	I **Reverse Yield Factor**	J **Allocation Factor**	K **Preparation process code**	L **Preparation process**	M **Preparation reverse yield**	N **Include ingredient's preparation (1/0)**
A.11.04.017	Sunflower oil	A.04.04.005	Sunflower seed (Helianthus annuus)	F28.A0C02$ F02.A068M	PROCESS=Oil production, PART-NATURE=Vegetable fats and oils (as part-nature)	100	1	**2.44**	**0.82**				1

1. Energy file: Oil production = **1.027 MJ per kg seed**.
2. Reverse yield **2.44** kg seed per kg oil; allocation **0.82** (the meal carries the rest).
3. Per kg oil: 1.027 × 2.44 × 0.82 = **2.054 MJ**.
4. In the bread: 0.0286 kg oil × 2.054 = **0.059 MJ**.

Water, salt, yeast. Their rows say Unspecified and they have no preparation process, so they add **0**.

Step 2: preparation energy (baking)

Baking adds **3.27 MJ** per kg bread. The preparation process is counted **once for the food**, not per ingredient.

1. Energy file: Roasting (used as the proxy for baking) = **2.921 MJ per kg raw material** going into the oven.
2. Preparation reverse yield **1.12**: about 1.12 kg dough goes into the oven per kg bread that comes out (10–12 % water evaporates).
3. Per kg bread: 2.921 × 1.12 = **3.27 MJ**.

The preparation reverse yield is what keeps the per-kg-raw value correct when it is applied to 1 kg of finished food. It matters most where much water is lost (crispbread 1.4, dried sausage up to 1.5, corn chips 1.4).

Step 3: total and energy carriers

1 kg white wheat bread = **3.62 MJ**, of which 0.69 MJ electricity and 2.92 MJ natural gas. Each process brings its own carrier split from the energy file, applied to its MJ.

Part	MJ per kg bread	Electricity (MJ)	Natural gas (MJ)	Other (MJ)
Flour: grain milling	0.284	0.284 (100 %)	–	–
Oil: oil production	0.059	0.010 (17 %)	0.049 (83 %)	0.0002 heating oil
Baking (Roasting)	3.272	0.398 (12 %)	2.874 (88 %)	–
Total	**3.615**	**0.692**	**2.923**	**0.0002**

The model then turns each carrier's MJ into emissions with that carrier's emission factor.

Example 2: Oat porridge with milk

Oat porridge (A.01.06.007.001) needs **1.55 MJ/kg made with milk and 1.24 MJ/kg made with water**. It shows two more rules: alternatives with a probability, and when an ingredient's own preparation comes along.

Ingredient	kg per kg porridge	Probability	Process on the row
Oat flakes	0.15	1	none (own recipe: oats, Flattening / rolling)
Whole milk	0.85	0.5	Unspecified
Tap water	0.85	0.5	Unspecified
Porridge itself	**1**	**–**	**preparation: Cooking, preparation reverse yield 1**

The recipes as they stand in the recipe file, all columns: the porridge, then its two ingredients that have their own recipe. Oat flakes have no preparation process; whole milk has Market milk processing.

FOOD		INGREDIENT (COMPONENT)		PROCESS ON THE ROW		AMOUNT AND FACTORS				FOOD'S PREPARATION (SAME ON EVERY ROW, COUNTED ONCE)			
A **Food code**	B **Food name**	C **Component code**	D **Component name**	E **Facet(s) code**	F **Facet(s) Description**	G **Percentage**	H **Probability**	I **Reverse Yield Factor**	J **Allocation Factor**	K **Preparation process code**	L **Preparation process**	M **Preparation reverse yield**	N **Include ingredient's preparation (1/0)**
A.01.06.007.001	Oat porridge	A.01.06.001.011	Oat flakes	To be further disaggregated (d)		15	1	1	1	F28.A0BA1	Cooking and similar thermal preparation processes	1	1
A.01.06.007.001	Oat porridge	A.08.01.001.002	Cow milk, 3 - 4% fat (whole milk)	F28.A07XD	PROCESS=Unspecified	85	0.5	1	1	F28.A0BA1	Cooking and similar thermal preparation processes	1	1
A.01.06.007.001	Oat porridge	A.15.01	Tap water	F28.A07XD	PROCESS=Unspecified	85	0.5	1	1	F28.A0BA1	Cooking and similar thermal preparation processes	1	1
A.01.06.001.011	Oat flakes	A.01.02.008	Oats, grain	F28.A07LH	PROCESS=Flattening / rolling	100	1	**1.56**	**0.91**				1
A.08.01.001.002	Cow milk, 3 - 4% fat (whole milk)	A.08.01.001.002	Cow milk, 3 - 4% fat (whole milk)	F28.A07XD	PROCESS=Unspecified	100	1	1	1	F28.NEW09	Market milk processing (whole dairy: drinking milk, cream)	1	1

Milk and water are **alternatives** (probability 0.5 each): the porridge is made with one or the other, not both.

1. **Oat flakes:** 0.623 MJ per kg oats × 1.56 kg oats per kg flakes × allocation 0.91 = 0.884 MJ per kg flakes; × 0.15 kg = **0.133 MJ**.
2. **Milk:** the row says Unspecified, so this is bought drinking milk and its own preparation comes along: Market milk processing (whole dairy) 0.36 MJ per kg × 0.85 kg = **0.306 MJ**.
   The whole-milk recipe in the recipe file: milk is its own raw commodity (Unspecified, reverse yield 1, allocation 1), and its energy is only the preparation process Market milk processing.

FOOD		INGREDIENT (COMPONENT)		PROCESS ON THE ROW		AMOUNT AND FACTORS				FOOD'S PREPARATION (SAME ON EVERY ROW, COUNTED ONCE)			
A **Food code**	B **Food name**	C **Component code**	D **Component name**	E **Facet(s) code**	F **Facet(s) Description**	G **Percentage**	H **Probability**	I **Reverse Yield Factor**	J **Allocation Factor**	K **Preparation process code**	L **Preparation process**	M **Preparation reverse yield**	N **Include ingredient's preparation (1/0)**
A.08.01.001.002	Cow milk, 3 - 4% fat (whole milk)	A.08.01.001.002	Cow milk, 3 - 4% fat (whole milk)	F28.A07XD	PROCESS=Unspecified	100	1	1	1	F28.NEW09	Market milk processing (whole dairy: drinking milk, cream)	1	1

3. **Water: 0**.
4. **Cooking the porridge:** 1.108 MJ per kg × 1 = **1.108 MJ**.

	Made with milk	Made with water
Oat flakes	0.133	0.133
Milk (market-milk dairy)	0.306	–
Cooking	1.108	1.108
Total MJ per kg porridge	**1.547**	**1.241**

When an ingredient's preparation comes along. It does, unless the row's column N (Include ingredient's preparation) is 0. N is 0 only on rows with Cheesemaking, Fermentation, Churning, Separation or Drying of a dairy ingredient: there the dairy energy is already counted, either in the row's whole-plant value or in the product's own market-milk preparation. Everywhere else N is 1. Milk in porridge, cakes or sauces is bought milk, so it brings market-milk energy. Milk in cheese has the row process Cheesemaking: the cheese-dairy value (0.576 MJ per kg milk) already covers the whole plant, so the market-milk energy does not come along.

Example, Cheddar in the recipe file: the whole-milk row has the process Cheesemaking, so the milk's own preparation (Market milk processing) does not come along. Energy per kg Cheddar = 0.576 MJ per kg milk × 8.7 kg milk × allocation 0.94 = **4.71 MJ**.

FOOD		INGREDIENT (COMPONENT)		PROCESS ON THE ROW		AMOUNT AND FACTORS				FOOD'S PREPARATION (SAME ON EVERY ROW, COUNTED ONCE)			
A **Food code**	B **Food name**	C **Component code**	D **Component name**	E **Facet(s) code**	F **Facet(s) Description**	G **Percentage**	H **Probability**	I **Reverse Yield Factor**	J **Allocation Factor**	K **Preparation process code**	L **Preparation process**	M **Preparation reverse yield**	N **Include ingredient's preparation (1/0)**
A.08.08.049	Cheese, Cheddar	A.08.01.001.002	Cow milk, 3 - 4% fat (whole milk)	F28.A0C6E$ F02.A06EX$ F07.A073T	PROCESS=Cheesemaking, PART-NATURE=Cheese (as part-nature), FAT-CONTENT=30 % fat	100	1	**8.7**	**0.94**				**0**

Example 3: Pizza, cheese (preparation counted only partly)

1 kg of cheese pizza (I.19.01.002.013) needs **4.58 MJ** of processing energy. It shows the share rule for column M: the pizza base is already baked in its own recipe, so baking the pizza counts only for the part that has not been baked yet (the topping).

Ingredient	kg per kg pizza	MJ per kg ingredient	MJ per kg pizza
Pizza base, cooked	0.564	4.068 (of which 3.272 its own baking)	2.294
Cheese, Mozzarella	0.211	2.983 (cheesemaking, see Example 4)	0.629
Tomato purée	0.211	1.193 (pulping + concentration)	0.252
Salt	0.014	0	0
Baking the pizza (preparation, M 0.48)	–	2.921 × 0.48	1.402
Total			**4.578**

The pizza recipe as it stands in the recipe file, all columns:

FOOD		INGREDIENT (COMPONENT)		PROCESS ON THE ROW		AMOUNT AND FACTORS				FOOD'S PREPARATION (SAME ON EVERY ROW, COUNTED ONCE)			
A **Food code**	B **Food name**	C **Component code**	D **Component name**	E **Facet(s) code**	F **Facet(s) Description**	G **Percentage**	H **Probability**	I **Reverse Yield Factor**	J **Allocation Factor**	K **Preparation process code**	L **Preparation process**	M **Preparation reverse yield**	N **Include ingredient's preparation (1/0)**
I.19.01.002.013	Pizza, cheese	A.02.14.003	Tomato purée	To be further disaggregated (d)		21.1	1	1	1	F28.A07GY	Roasting	0.48	1
I.19.01.002.013	Pizza, cheese	A.08.08.120	Cheese, Mozzarella	To be further disaggregated (d)		21.1	1	1	1	F28.A07GY	Roasting	0.48	1
I.19.01.002.013	Pizza, cheese	A.16.04.001	Salt	F28.A07XD	PROCESS=Unspecified	1.4	1	1	1	F28.A07GY	Roasting	0.48	1
I.19.01.002.013	Pizza, cheese	I.19.01.002.009	Pizza base, cooked	To be further disaggregated (d)		56.4	1	1	1	F28.A07GY	Roasting	0.48	1

The pizza base has its own recipe, with its own baking (Roasting, M 1.12, like bread):

FOOD		INGREDIENT (COMPONENT)		PROCESS ON THE ROW		AMOUNT AND FACTORS				FOOD'S PREPARATION (SAME ON EVERY ROW, COUNTED ONCE)			
A Food code	B Food name	C Component code	D Component name	E Facet(s) code	F Facet(s) Description	G Percentage	H Probability	I Reverse Yield Factor	J Allocation Factor	K Preparation process code	L Preparation process	M Preparation reverse yield	N Include ingredient's preparation (1/0)
I.19.01.002.009	Pizza base, cooked	A.01.03.001.003	Wheat flour, white	To be further disaggregated (d)		65.84	1	1	1	F28.A07GY	Roasting	1.12	1
I.19.01.002.009	Pizza base, cooked	A.10.01.001	White sugar	To be further disaggregated (d)		2.08	1	1	1	F28.A07GY	Roasting	1.12	1
I.19.01.002.009	Pizza base, cooked	A.11.04.008	Olive oil	To be further disaggregated (d)		8.83	1	1	1	F28.A07GY	Roasting	1.12	1
I.19.01.002.009	Pizza base, cooked	A.15.01	Tap water	F28.A07XD	PROCESS=Unspecified	20	1	1	1	F28.A07GY	Roasting	1.12	1
I.19.01.002.009	Pizza base, cooked	A.16.04.001	Salt	F28.A07XD	PROCESS=Unspecified	1.3	1	1	1	F28.A07GY	Roasting	1.12	1
I.19.01.002.009	Pizza base, cooked	A.16.04.015	Yeast extract	To be further disaggregated (d)		1.95	1	1	1	F28.A07GY	Roasting	1.12	1

- **Pizza base, per kg base:** flour 0.6584 × 0.383 = 0.252, sugar 0.0208 × 6.863 = 0.143, olive oil 0.0883 × 4.363 = 0.385, yeast extract 0.0195 × 0.811 = 0.016, water and salt 0, baking 2.921 × 1.12 = 3.272. Total 4.068 MJ per kg base. Column N is 1 on the pizza's base row, so the base brings its baking along: 0.564 × 4.068 = 2.294 MJ per kg pizza.
- **Why M is 0.48 and not 1.1.** M = weight-loss factor × (1 − share already baked). The weight-loss factor for baking a pizza is 1.1. Share already through the same process (Roasting): the base, 0.564, because its own preparation is Roasting and it comes along (N = 1). Mozzarella (Cheesemaking) and tomato purée (Pulping, Concentration) have not been baked. M = 1.1 × (1 − 0.564) = 0.48.
- **What the rule prevents.** With M = 1.1 the pizza's baking would be 2.921 × 1.1 = 3.21 MJ instead of 1.40, and the base would be baked twice (its own 3.272 per kg already sits in the 2.294). Total 6.39 instead of 4.58 MJ per kg.

Example 4: Cheese, Mozzarella (the cheese dairy replaces the milk's preparation)

1 kg of mozzarella (A.08.08.120), the cheese on the pizza above, needs **2.98 MJ**. It shows column N = 0: the milk's own preparation (Market milk processing) is not added, because the cheese dairy value already covers the whole plant.

FOOD		INGREDIENT (COMPONENT)		PROCESS ON THE ROW		AMOUNT AND FACTORS				FOOD'S PREPARATION (SAME ON EVERY ROW, COUNTED ONCE)			
A **Food code**	B **Food name**	C **Component code**	D **Component name**	E **Facet(s) code**	F **Facet(s) Description**	G **Percentage**	H **Probability**	I **Reverse Yield Factor**	J **Allocation Factor**	K **Preparation process code**	L **Preparation process**	M **Preparation reverse yield**	N **Include ingredient's preparation (1/0)**
A.08.08.120	Cheese, Mozzarella	A.08.01.001.002	Cow milk, 3 - 4% fat (whole milk)	F28.A0C6E$ F02.A06EX$ F07.A073G	PROCESS=Cheesemaking, PART-NATURE=Cheese (as part-nature), FAT-CONTENT=19 % fat	100	1	**5.51**	**0.94**				**0**

The whole-milk recipe is the one shown in Example 2: milk is its own raw commodity, and its only energy is its preparation, Market milk processing 0.36 MJ per kg.

- **Row process:** Cheesemaking = 0.576 MJ per kg milk (JRC whole-plant value for a cheese dairy: milk reception, pasteurising, cheesemaking, cleaning).
- **The milk's own energy:** whole milk has no processing of its own on the way from the farm; it is a raw commodity. Its only energy is its preparation step, Market milk processing: 0.36 MJ per kg, the energy of a drinking-milk dairy. Normally that step comes along when milk is an ingredient (as in the porridge). Here column N is 0, so the model leaves it out: it takes the milk's energy (0.36) and subtracts its preparation (0.36), which leaves 0. Reason: the cheese dairy value on the row (0.576) already includes receiving and pasteurising the milk, so adding the drinking-milk dairy as well would count those steps twice.
- **Per kg mozzarella:** 5.51 kg milk × allocation 0.94 × (0.576 + 0) = 2.983 MJ.
- **If N were 1:** 5.51 × 0.94 × (0.576 + 0.36) = 4.85 MJ. The market-milk dairy would be counted on top of the cheese dairy, which already does the same steps.
- **Compare with Example 2:** milk in porridge has the row process Unspecified and N = 1. That is bought drinking milk, so its market-milk energy does come along.

- **In the pizza:** mozzarella has no preparation of its own, so its 2.983 MJ per kg comes along whole (0.211 × 2.983 = 0.629 MJ per kg pizza).

Rules and sources

1. Energy values are **per kg raw material entering the process**, never per kg product; the reverse yield converts them to the food.
2. **Allocation** shares a process's burden between co-products (flour/bran/germ, oil/meal, cream/skimmed milk); the factors for one process add up to 1.
3. Several processes on one row are **added up** (for example Drying + Grinding).
4. Some foods have **one preparation process** (never more than one; many, such as flour, oil and cheese, have none), counted once per kg food × its preparation reverse yield (column M). M = weight-loss factor × (1 − share of the food already through the same process, through all recipe levels): fruit tart 1.1 × (1 − 0.353 pastry) = 0.71, bread with seeds 1.12 × (1 − 0.899 bread) = 0.11, a food fully made of prepared ingredients gets none.
5. A food used as an ingredient brings its own recipe and preparation along (bread inside bread with seeds), **unless the row's column N (Include ingredient's preparation) is 0. It is 0 only on rows with Cheesemaking, Fermentation, Churning, Separation or Drying of a dairy ingredient, where the dairy energy is already counted**.
6. Alternatives in a recipe carry a **probability**; the shares are per alternative, not added together.

Main sources: JRC BREF Food, Drink and Milk Industries (Santonja et al. 2019) for most process values; the tab Documentation & sources in the energy file for each value; limitations L1–L48 in the tab Limitations. Changes to recipes, reverse yields and allocation are listed in the tab Changes of the recipe file.

How to implement this in code

The model needs no food-specific exceptions: every decision is a value in columns K–N of tab Recipes. It reads two inputs and applies one recursive function.

Inputs

- Energy file: one value per process code (`F28.xxxxx`), MJ final energy per kg raw material, plus its split into energy carriers.
- Recipe file, tab Recipes: one row per ingredient, 14 columns (A Food code … N Include ingredient's preparation). Group the rows by Food code.

Steps

1. **Process codes.** Columns E (Facet(s) code) and K (Preparation process code) can hold several codes joined with `$`. Split on `$` and keep only codes that start with `F28.`; the others (F02, F07, F10 …) describe the product and carry no energy. Several processes on one row are added up.
2. **Preparation of a food:** prep(f) = Σ C(codes in column K) × column M. Read it once per food (from its first row; it is the same on every row). Blank K → 0.
3. **Energy of a food, recursive:** for every row i of food f: weight = Percentage / 100 × Probability × Reverse Yield Factor × Allocation Factor; row energy = Σ C(F28 codes in column E). If the component is itself a food in the recipe file (and not the food itself), add its energy E(component); if column N is 0, subtract its preparation prep(component). Add weight × (row energy + component energy). After the rows, add prep(f) once.
4. **A component that is the food itself** (raw commodity, for example whole milk made of whole milk) has no nested energy: only its row process and its own preparation.
5. **Energy carriers:** run the same calculation with a vector per process (electricity, natural gas, …) instead of one number, and add vectors. Each carrier is then multiplied by its emission factor.
6. **Guards:** memoise E(f) (foods are reused many times); stop on cycles (a food that reaches itself through its ingredients); report missing process codes instead of silently using 0; check that Σ Percentage × Probability = 100 for each food.

Do not: add the preparation per row (it repeats on every row only for readability); recalculate M (it is already in the file, from prep_rules.py); add exceptions for dairy, bread or pizza (they are all in columns M and N).

```
def f28(cell):
    return [c for c in str(cell).split('$') if c.startswith('F28.')] if cell else []

def prep(food):
    r = recipes[food][0]                        # same on every row
    return sum(C[c] for c in f28(r.prep_code)) * (r.prep_ry or 0)

@cache
def E(food):
    total = 0.0
    for r in recipes[food]:
        w = r.percentage / 100 * r.probability * r.reverse_yield * r.allocation
        row = sum(C[c] for c in f28(r.facet_code))
        nested = 0.0
        if r.component != food and r.component in recipes:
            nested = E(r.component)
            if r.include_prep == 0:
                nested -= prep(r.component)
        total += w * (row + nested)
    return total + prep(food)
```

Test values (MJ per kg, recipe and energy files of 3 Nov 2026): Wheat bread, white 3.615; Oat porridge 1.394 (average of milk 1.547 and water 1.241, probability 0.5 each); Cheese, Cheddar 4.711; Cheese, Mozzarella 2.983; Pizza base, cooked 4.068; Pizza, cheese 4.578. A working reference implementation is energy_model.py (Python, reads the two Excel files).